KEY PERFORMANCE INDICATORS (KPIs)

Banking Transaction System Software Testing Track

Project scope: A Java banking system that manages accounts, deposits, withdrawals, and transaction history, with exception handling, file/database storage, user interface/API layers, and security mechanisms. Testing activities include manual testing, static testing and Jira feedback, Selenium UI automation, Postman API testing, security testing, performance testing, .and usability verification

How to use this document: The indicators below are proposed measures for the team. The numeric targets are suggested starting points—not requirements stated in the project .brief—and should be adjusted to the agreed scope and schedule

1. Functional Testing KPIs

- **Core workflow test coverage:** Percentage of planned test cases covering account operations, deposits, withdrawals, and transaction history. Suggested target: 100% of in-scope requirements covered.
- **Test execution progress:** Number of test cases executed divided by the total planned test cases. Suggested target: 100% of planned tests executed before sign-off.
- **Test pass rate:** Number of passed test cases divided by the number of executed test cases x 100. Suggested target: At least 95% before release, with no unresolved critical defects.
- **Requirement traceability:** Percentage of in-scope requirements linked to one or more test cases. Suggested target: 100%.

2. Transaction Accuracy and Validation KPIs

- **Transaction calculation accuracy:** Percentage of tested deposits and withdrawals that produce the expected account balance. Suggested target: 100% of tested transactions match expected results.
- **Invalid transaction rejection rate:** Percentage of invalid transaction scenarios correctly rejected, such as withdrawals exceeding available balance or invalid amounts. Suggested target: 100% of defined negative tests behave as expected.
- **Transaction history accuracy:** Percentage of tested transactions correctly recorded with the expected type, amount, timestamp, and resulting balance/history details. Suggested target: 100% of tested records are accurate.
- **Exception-handling effectiveness:** Percentage of defined error scenarios handled safely without an unexpected crash or incorrect account update. Suggested target: 100% of defined exception scenarios pass.

3. Performance and Load Testing KPIs

- **Response time compliance:** Percentage of critical transactions (deposits, withdrawals, balance queries) completed within agreed time limits (e.g., < 1.0 second). Suggested target: 95% of tested transactions meet response time thresholds.
- **Concurrent transaction handling:** Verification of system behavior during simultaneous transaction requests to prevent race conditions or balance discrepancies. Suggested target: 100% data integrity maintained under concurrent operations.
- **Resource utilization stability:** System stability and memory leak monitoring during sustained operational testing runs. Suggested target: Zero unhandled memory leaks or severe resource exhaustion during prolonged execution.
- **Throughput capacity (TPS):** Measurement of successful transactions processed per second under nominal and peak load conditions. Suggested target: Maintain target TPS without drop in transaction success rate.
- **System recovery time post-peak:** Duration required for system resources (CPU, RAM, thread pools) to normalize after a sudden burst of transaction load. Suggested target: Recovery to baseline within 30 seconds post-burst.

4. Security and Access Control KPIs

- **Authentication & authorization coverage:** Percentage of tested user roles (Customer, Admin) verified against improper access or privilege escalation attempts. Suggested target: 100% of security rules validated with zero unauthorized access vulnerabilities.
- **Input validation & injection prevention:** Percentage of input fields (account numbers, amounts, user IDs) tested against malicious scripts, SQL injections, or boundary overflow inputs. Suggested target: 100% of invalid/malicious inputs safely rejected.
- **Data privacy & masking:** Verification that sensitive banking data (passwords, complete account numbers) are properly masked in logs and user interfaces. Suggested target: 100% compliance with data privacy guidelines.
- **Session management integrity:** Evaluation of session timeouts, token invalidation upon logout, and resistance to session hijacking or replay attacks. Suggested target: 100% compliance with secure session handling.
- **Audit trail completeness:** Verification that security-relevant events (failed logins, privilege changes, administrative overrides) are immutably logged with accurate metadata. Suggested target: 100% logging coverage of defined audit events.

5. Defect Management KPIs

- **Defect density:** Number of confirmed defects found per tested feature or module; track by module to identify areas needing extra attention.
- **Defect severity distribution:** Count of open and closed defects by severity (critical, high, medium, low) to communicate project risk.
- **Defect leakage:** Number of defects discovered after a test phase or sign-off that should have been detected earlier. Suggested target: Zero known critical defects escaping final testing.

- **Defect retest success rate:** Percentage of fixed defects that pass retesting. Track reopened defects separately to identify incomplete fixes.

6. Automation and API Testing KPIs

- **UI automation coverage:** Percentage of selected high-priority UI workflows automated with Selenium WebDriver. Suggested target: Prioritize core account and transaction workflows.
- **Automation pass rate:** Percentage of executed automated tests that pass. Investigate failures to distinguish product defects from unstable tests or environment issues.
- **API test coverage (if applicable):** Percentage of in-scope backend endpoints and scenarios tested with Postman, including valid requests, HTTP status code validations (200, 400, 401, 403, 500), and negative/error cases.
- **Automated test stability:** Percentage of repeated automation runs that produce consistent results in the same environment.

7. Data Persistence and Reliability KPIs

- **Data persistence success rate:** Percentage of tested transactions that remain correctly stored and retrievable after the relevant save/reload process or system restart.
- **Data consistency:** Percentage of tested cases where the account balance and transaction history remain consistent after deposits, withdrawals, and repeated operations.
- **Recovery from failed operations:** Percentage of tested failure scenarios where the system avoids saving partial or incorrect transaction data, ensuring transaction rollback where required.

8. Usability and User Experience KPIs

- **User message clarity:** Percentage of user-facing error messages that clearly state the cause and solution without exposing raw code exceptions or technical stack traces. Suggested target: 100% clear, user-friendly messages.
- **Interface workflow consistency:** Verification that navigation, input labels, and confirmation dialogs follow standard, consistent banking interface patterns. Suggested target: 100% consistency across all modules.
- **Task completion rate:** Percentage of users or testers able to complete core workflows (e.g., executing a fund transfer, generating a statement) without external guidance. Suggested target: >= 95% completion on first attempt.
- **Accessibility compliance:** Verification of key interface elements against standard accessibility guidelines (e.g., keyboard navigation, high-contrast visibility, screen reader tags). Suggested target: 100% compliance on primary customer interfaces.

9. Jira and Testing Process KPIs

- **Test and defect tracking completeness:** Percentage of executed tests and confirmed defects recorded in Jira with status and relevant details. Suggested target: 100%.
- **Defect turnaround time:** Average time from defect reporting to resolution; review by severity and agreed team workflow.
- **Review feedback closure:** Percentage of actionable static-testing/code-review findings that are addressed or formally dispositioned.
- **Test reporting cadence:** Publish an agreed progress summary showing executed/passed/failed/blocked tests, open defects by severity, and key risks.

10. KPI Summary Table

KPI	How to Measure	Suggested Target / Goal
Requirement coverage	Requirements with linked test cases / total in-scope requirements x 100	100%
Test execution	Executed test cases / planned test cases x 100	100% before sign-off
Test pass rate	Passed test cases / executed test cases x 100	>= 95%, with no open critical defects
Transaction accuracy	Transactions matching expected outcomes / transactions tested x 100	100% of tested cases
Negative-test handling	Defined invalid scenarios handled correctly / invalid scenarios tested x 100	100% of defined cases
Security & access control	Access rule scenarios passed / total access scenarios tested x 100	100%
Response time compliance	Transactions within threshold / total performance tests x 100	>= 95%
Throughput capacity	Peak successful operations processed per second / target TPS x 100	100% of target load
Audit trail completeness	Logged security events / total triggered audit events x 100	100%

KPI	How to Measure	Suggested Target / Goal
Task completion rate	Successful unassisted workflow completions / total usability trials x 100	>= 95%
Jira tracking completeness	Tests and defects recorded correctly / total executed tests & confirmed defects x 100	100%

11. Reporting and Review

- Review the KPIs at each agreed testing checkpoint and update the values using actual Jira/test execution data.
- Report the measurement period, numerator, denominator, and any exclusions so the metrics can be understood consistently.
- Do not treat a high pass rate alone as proof of quality; consider test coverage, defect severity, transaction accuracy, performance stability, security validation, and unresolved risks together.
- Mark UI automation, API testing, or performance tools as not applicable if the corresponding interface or backend endpoints are not implemented in the project scope.